

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	06 May 2023
Team ID	NM2023TMID10056
Project Name	AI Enabled Car Parking Using Open CV

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Vehicle Detection	a. Detect vehicles in real-time using video or image inputs. b. Identify and differentiate between different types of vehicles (cars, motorcycles, trucks, etc.). c. Handle different lighting conditions and environmental factors for accurate detection.
FR-2	License Plate Recognition	a. Extract and recognize license plate characters accurately. b. Handle variations in license plate designs and formats. c. Ensure reliable recognition under different lighting and weather conditions.
FR-3	Parking Space Detection	a. Identify empty parking spaces within the parking area. b. Distinguish between occupied and available parking spaces. c. Handle complex parking lot layouts and obstacles accurately.
FR-4	Parking Space Allocation	a. Assign the nearest available parking space to a detected vehicle. b. Consider the size and type of vehicle for appropriate space allocation. c. Optimize the allocation process to maximize space utilization.
FR-5	Parking Space Reservation	a. Allow users to reserve parking spaces through a user-friendly interface. b. Update parking space availability in real-time based on reservations.

		c. Provide confirmation and reservation details to the user.
FR-6	Vehicle Tracking	a. Track the movement of vehicles within the parking area accurately. b. Update the status of occupied and vacant parking spaces based on vehicle movement. c. Provide real-time notifications for any unauthorized or suspicious activities.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have an intuitive and user-friendly interface that is easy to understand and navigate, ensuring a positive user experience for both drivers and parking administrators.
NFR-2	Security	The system should ensure the security and privacy of user data, including license plate information and payment details. It should implement appropriate encryption measures and access controls to protect sensitive information.
NFR-3	Reliability	The system should be highly reliable, minimizing system failures and errors. It should be able to handle exceptions gracefully and recover from any unexpected failures without compromising the overall functionality.
NFR-4	Performance	The system should define and measure performance metrics, such as vehicle detection accuracy, processing speed, and user satisfaction, to continuously monitor and improve its performance over time.
NFR-5	Availability	The system should have a high level of availability, ensuring that it is accessible and operational for users at all times, with minimal downtime or scheduled

		maintenance windows.
NFR-6	Scalability	The system should be scalable to accommodate parking lots of varying sizes, from small parking areas to large multi-level parking structures, without sacrificing performance or accuracy.